Journal of Nuclear Materials 426 (2012) 52–58

Contents lists available at SciVerse ScienceDirect

Journal of Nuclear Materials

journal homepage: www.elsevier.com/locate/jnucmat

Quantitative characterization of microstructural defects in up to 32 dpa neutron irradiated EUROFER97

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ARTICLE INFO

Article history:
Received 6 July 2011
Accepted 16 March 2012
Available online 28 March 2012

ABSTRACT

The microstructure of the neutron-irradiated reduced activation ferritic/martensitic (RAFM) steel EUROFER97 was evaluated by transmission electron microscopy (TEM). Emphasis was put on analyzing the influence of the irradiation dose on the evolution of size and density of microstructural defects like dislocation loops and voids at low irradiation temperatures of 330–340 °C. To study the dose dependence, samples irradiated to 15 and 32 dpa were analyzed. The weak-beam dark-field (WBDF) technique was applied to analyze dislocation loops and small defect clusters using different diffraction conditions. The average densities and sizes of the defects increase slightly from 1.4×10^{22} m^{-3} and 3.4 nm at 15 dpa to 1.7×10^{22} m^{-3} and 4.8 nm at 32 dpa. Through-focus series also revealed the presence of small voids in the material, but with a density at least one order of magnitude lower than that of the dislocation loops. In order to correlate the irradiation induced changes in the microstructure to the changes in the mechanical properties, the obtained quantitative data was used to estimate dose-dependent hardening with the dispersed barrier hardening model. The estimation of hardening by using recent literature results on the loop obstacle strength shows, that alone the defects visible in the TEM are not sufficient to explain the hardening quantified in the post-irradiation tensile tests.

1. Introduction

Reduced activation ferritic/martensitic (RAFM) steels like EUROFER97 are primary candidate structural materials for in-vessel components of future fusion power plants and therefore will be exposed to high neutron and thermo-mechanical loads. When impinging the material, the fusion neutrons create atomic displacement cascades, and besides a considerable amount of helium is generated by transmutation. The displacement damage accumulates during reactor operation so that point-defect clusters and dislocation loops are created. In conjunction with He-bubbles and radiation induced precipitates these defects lead to strong low temperature hardening and embrittlement of the material [1].

Aim of this work is quantifying the influence of neutron irradiation on the microstructure of RAFM steels. Since no irradiation facility with a fusion-like neutron spectrum exists, steel samples were irradiated under fission conditions to simulate fusion neutron damage. The samples were then investigated by transmission electron microscopy (TEM). Emphasis was put on measuring the sizes and volume densities of irradiation-induced defects. These irradiation induced changes in the microstructure should be correlated to the changes in the mechanical material properties by using appropriate models.

2. Experimental

Rolled plate material of EUROFER97 (Heat 83697) with a composition of 8.91Cr 1.08W 0.48Mn 0.20V 0.14Ta 0.006Ti 0.12C (wt.%, Fe balance) [2] was produced by Böhler Austria GmbH. The steel was austenitized at 960 °C for 31 min and tempered at 760 °C for 90 min in air by the manufacturer. To study the irradiation performance miniaturized Charpy impact, tensile and LCF specimens of EUROFER97 and other RAFM steels were irradiated to doses of 15 and 32 dpa in the BOR 60 fast reactor at Joint Stock Company (JSC) "State Scientific Centre Research Institute of Atomic Reactors" (SSC RIAR) in Dimitrovgrad, within the framework of the "Wissenschaftlich-Technische Zusammenarbeit 01/577" (WTZ) [3] and "Associated Reactor Irradiation in BOR 60" (ARBOR-1) [4] irradiation programs, respectively. The irradiation temperatures were between 330 and 340 °C and the neutron flux was 1.8×10^{19} m^{-2} s^{-1} (>0.1 MeV). The post irradiation mechanical testing of the specimens from WTZ and ARBOR-1 irradiation program was performed in the material science laboratory of RIAR. The 15 dpa specimen from the WTZ program used for the current study was impact-tested in the lower shelf at 20 °C. The 32 dpa specimen from the ARBOR-1 program was impact tested in the transition region close

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http://dx.doi.org/10.1016/j.jnucmat.2012.03.027

to the upper shelf at 140 °C (ARBOR-1). Thereafter the broken parts of impact tested specimens were transported to the Fusion Material Laboratory (FML) of Karlsruhe Institute of Technology (KIT) for microstructural investigations. In order to avoid the influence of specimen deformation from impact testing on the microstructure, the TEM specimens were manufactured from undeformed parts of the Charpy specimens by cutting off 150 μm thick slices using a cutting wheel. The slices were then thinned by electrolytic polishing in a solution of 20% H_2SO_4 + 80% CH_3OH at room temperature with a Tenupol-5 jet polisher. In order to minimize radioactivity and magnetism, discs of 1 mm diameter including the electron-transparent region were punched out and put in foldable copper nets for examination in the TEM. Modification of the microstructure due to the 1 mm disc punching was not observed. The TEM investigations were performed at 200 kV using a FEI Tecnai G^2 F20 microscope equipped with a post-column GIF Tridiem energy filter. The weak-beam dark-field (WBDF) technique [5] was used for imaging of radiation-induced defects. The diffraction conditions which were chosen are indicated in Section 3 and result in an excitation error s_g of approximately 0.2 nm^{-1}. All images were recorded with the GIF camera. To improve the contrast and remove contributions of inelastically scattered electrons in thicker regions of the samples the images were zero-loss filtered with an energy slit of 15 eV, see e.g. [6] for zero-loss filtering.

3. Results and discussion

3.1. Quantification of microstructural defects

A detailed bright-field image of the unirradiated EUROFER97 microstructure is shown in Fig. 1a. Except for line dislocations no features were visible. The martensitic lath structure was observable at lower magnifications (not shown), precipitates were decorating the lath boundaries. After irradiation to 15 dpa dislocation loops and small clusters visible as black dots could be identified, as shown in Fig. 1b. The density of defects seemed to be quite high inside this grain.

For quantitative analysis weak-beam dark-field imaging was used. The images of the defects were thereby confined to their physical sizes and narrow peaks allowed imaging of small defects down to 1 nm. At first a sample irradiated to 15 dpa was investigated. A representative grain was chosen and tilted to three different diffraction conditions near a $\langle 001 \rangle$ zone axis. Diffraction vectors of the $g = \{020\}$, $g = \{110\}$ and $g = \{130\}$ type were used to image microstructural defects. Two of the resulting micrographs are shown in Fig. 2a and b. Dislocation loops with diameters of up to 15 nm could be detected, as well as small defect clusters which are visible as individual white dots with high contrast to the background and thought to be of interstitial type. To some extend the identification of defects was hindered by the detrimental background contrast (i.e. the image contrast which could not be ascribed to dislocation loops or white dots) which probably stems from surface contamination of the sample.

Apart from dislocation loops and clusters, voids were found in the material. These defects were hardly visible in the focused image shown in Fig. 3a. When recording bright-field through-focal series the contrast of the defects changed from a dark dot with a white fresnel fringe in the overfocused image in Fig. 3b to a white dot with a black fringe in underfocus in Fig. 3c. This confirmed the presence of voids [7]. The voids in the investigated grain were counted and measured to get the size distribution displayed in Fig. 3d, the average size was 2.6 nm. The resulting volume density was 3.6×10^{20} m^{-3}, which is about two orders of magnitude below the density of loops in the same sample.

A preliminary analysis of a 32 dpa sample was presented in [8]. For the investigation two different grains were chosen. In order to enable subsequent Burgers vector determination grain #2 was tilted to five different diffraction conditions, which are indicated in Fig. 5. Two micrographs taken in grain #2 with *g (4.1 g)*, $g = \{200\}$ and $g\,(3.1\text{ g})$, $g = \{\bar{2}1\bar{1}\}$ near a $\langle 011 \rangle$ zone axis are shown in Fig. 4a and b. Here dislocation loops with diameters of up to 20 nm are visible. The loops generally appear edge on with a strong double arc contrast. By investigating the correlation between conventional TEM images of dislocation loops produced by Molecular Dynamic simulations in α-Fe and loop observation in TEM in neutron irradiated Fe–9Cr crystal it was shown in [9] that the appearance of double arc contrast is characteristic for interstitial type dislocation loops. Some defects were hard to identify, especially in Fig. 4b. This was on the one hand because it was not clear whether some white dots were actually small point defect clusters or belonged to one of the arcs of larger dislocation loops. On the other hand the identification of defects was hindered by the mentioned background contrast, which is more pronounced in Fig. 4b. In both micrographs it appears that most loops are aligned along {211} planes, which is common for bcc systems [10]. Also in samples irradiated to 32 dpa voids were detected. The density was

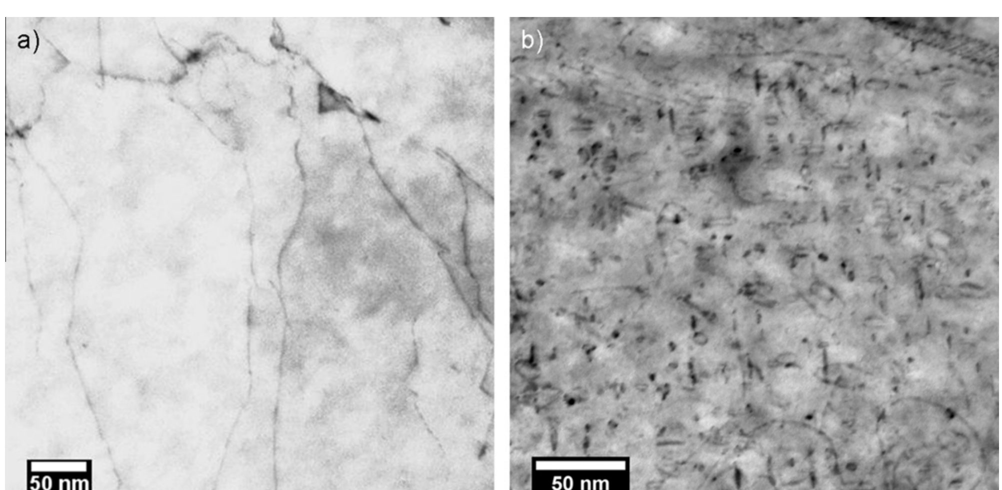

Fig. 1. (a) Bright-field image of unirradiated EUROFER97. Apart from line dislocations no microstructural features are visible. (b) Defect microstructure at slightly higher magnification showing dislocation loops, clusters (black dots) and line dislocations after irradiation to 15 dpa.

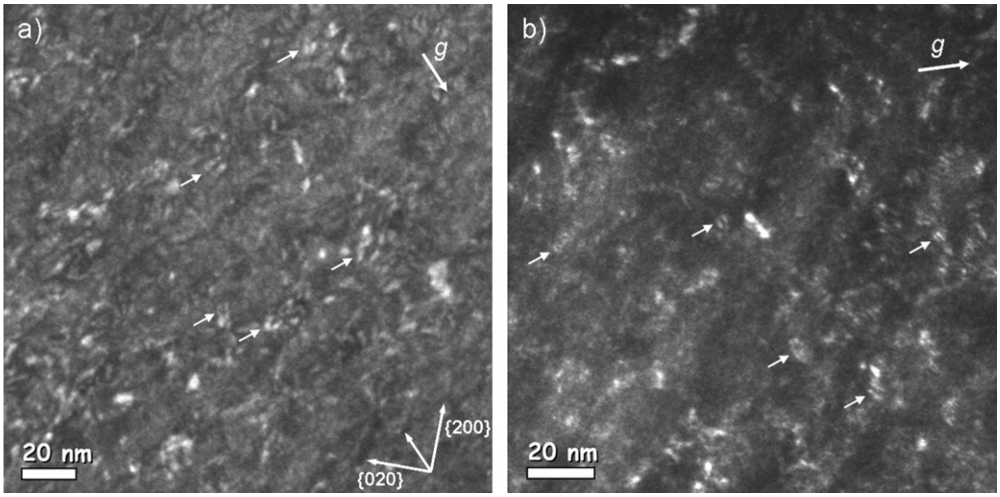

Fig. 2. WBDF micrographs of EUROFER97 irradiated to 15 dpa taken near a ⟨001⟩ zone axis: (a) with g (5.1 g), g = {110} and (b) with g (3.1 g), g = {130}. The visible dislocation loops have diameters of up to 15 nm. Some loops were marked with arrows for better visibility.

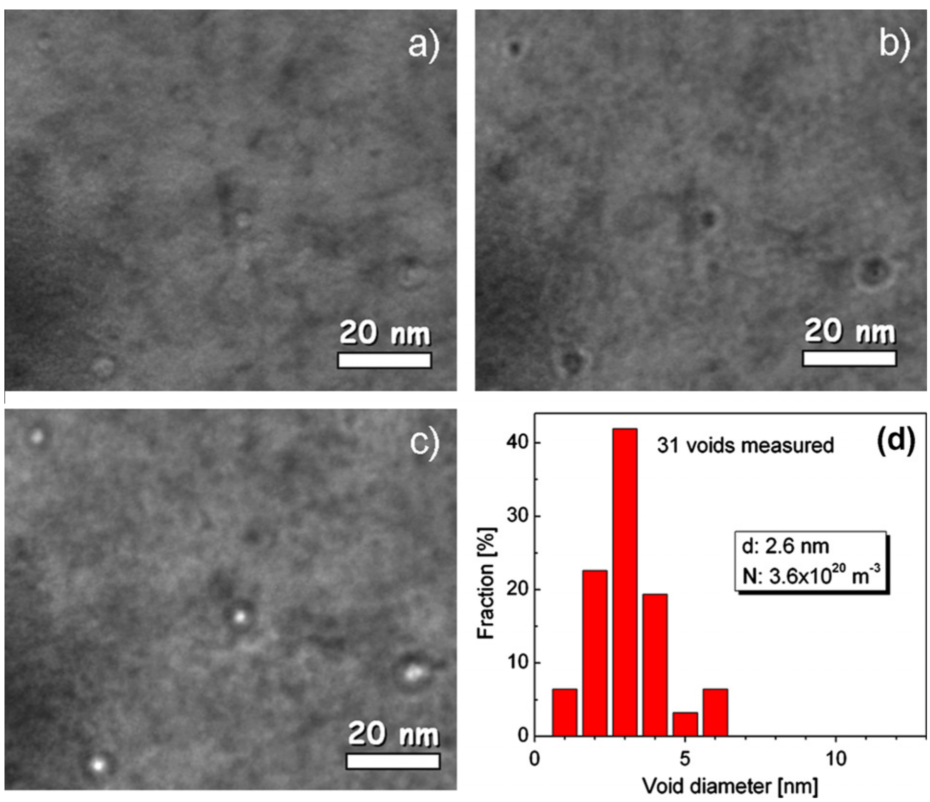

Fig. 3. Voids in a 15 dpa EUROFER97 sample. The kinematical BF-images were taken with g = {211} near a ⟨111⟩ zone axis: (a) focused image, voids are hardly visible, (b) overfocus, +2 µm, (c) underfocus, −2 µm and (d) volume density N, average size d and size distribution of voids in the investigated grain.

2.3×10^{21} m^{-3}, about one order of magnitude higher than observed in the 15 dpa sample, whereas the average size was 1.6 nm.

For determination of size distributions of defects we have not distinguished between dislocation loops and small clusters. After calibration of the TEM images the maximum diameters of the defects were marked manually with lines using the public domain software ImageJ [11]. Thereupon counting and sizing of the lines were done automatically with the built-in script functions. By this means areas of 0.75 µm^2 and 2.7 µm^2 of the 15 and 32 dpa sample were analyzed, respectively. To get the volume density of the defects the foil thickness was measured by convergent beam electron diffraction (CBED) according to Kelly et al. [12] and Allen [13]. For this purpose all investigated grains were tilted to different dynamical two-beam conditions. The recorded CBED patterns were analyzed using a DM-script by Hou [14] resulting in average thickness values of 175 nm for the grain in the 15 dpa sample and 146 nm and 106 nm for grain #1 and #2 in the 32 dpa sample. All size distributions of the defects are given in Fig. 5. The corresponding average sizes and volume densities of the defects are indicated in the graph of each diffraction condition. The maximum diameter of the visible dislocation loops was 20 nm for 15 dpa and 25 nm for 32 dpa, while the average diameter over all diffraction

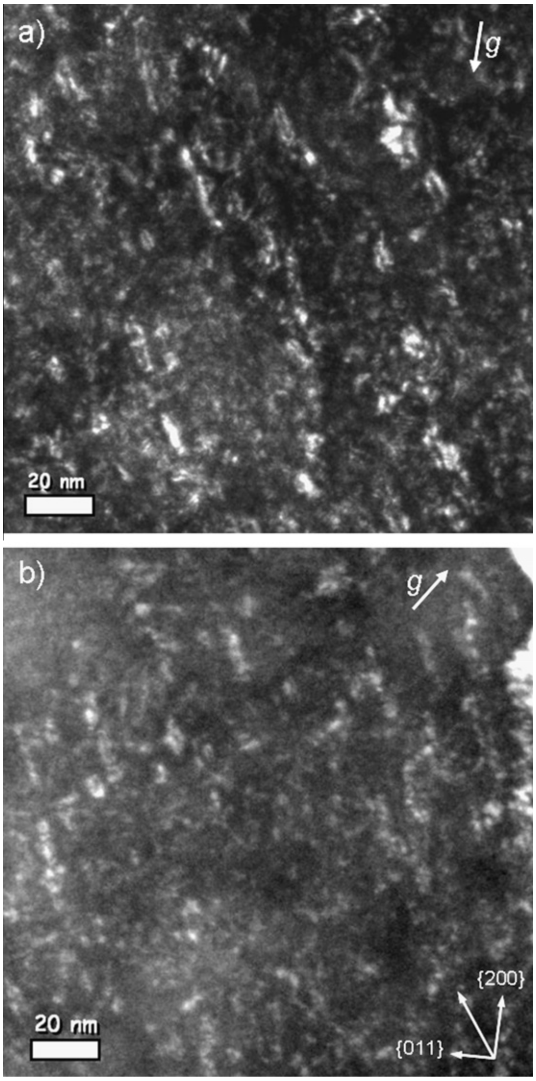

Fig. 4. Microstructural defects in EUROFER97 irradiated to 32 dpa. The WBDF images were taken near a $\langle 0\,1\,1\rangle$ zone axis: (a) with $g\,(4.1\,g)$, $g = \{2\,0\,0\}$ and (b) with $g\,(3.1\,g)$, $g = \{\bar{2}\,1\,\bar{1}\}$. The loops generally appear edge on with a strong double arc contrast and diameters of up to 20 nm.

conditions $\langle d\rangle$ increased from 3.4 nm to 4.8 nm. The average volume density $\langle N\rangle$ of the defects increased only slightly from $1.4 \times 10^{22}\ m^{-3}$ at 15 dpa to $1.7 \times 10^{22}\ m^{-3}$ at 32 dpa. In comparison to our earlier publication [8] the volume density measured in the 32 dpa sample is different, because for this work a lot more diffraction conditions were analyzed.

So far only very few quantitative microstructural results of irradiated EUROFER97 have been published [8,15]. Similarly to our results, loops with sizes between 5 and 25 nm were detected after irradiation to 16.3 dpa at 300 °C [15]. The defect density was about one order of magnitude lower than we measured, mainly because only defects clearly resolvable as dislocation loops were counted [16]. As listed in Table 1, investigations on F82H yielded densities similar to ours already at considerably lower irradiation doses of around 9 dpa. Unfortunately only very few consistent data sets concerning the dose-dependent density of neutron irradiation induced defects in RAFM steels are available. Corresponding results in [17] for example suggest, that the defect density first increases with the dose up to a dose between 2.5 and 8.8 dpa, and then decreases above that value. But since an influence of the irradiation temperature could not be excluded and the number of counted defects was rather low this dose-dependent behavior cannot be used to explain the relatively low defect densities we observed at 15 and 32 dpa. However, different metallurgical variables like the chemical composition, heat treatments and microstructure of EUROFER97 and F82H might influence defect accumulation during irradiation leading to different defect densities. Compared to the irradiation experiments listed in Table 1 the average sizes of the loops seem to be rather small for a high-dose irradiation. Possibly in our case the irradiation temperature was just below the threshold for loop coarsening, which is believed to lie in the range of 330 °C [17]. The size distribution of defects could be affected in two ways: First, in some cases it was not possible to identify large loops unambiguously because of complicated diffraction and background contrast. Also contrast of the arcs of some larger, but hardly visible loops could have been misleadingly counted as small clusters leading to an underestimation of size and density of larger defects.

3.2. Burgers vector determination

It is important to notice that due to different orientations with respect to the direction of the electron beam only a fraction of loops is visible in the analyzed micrographs. To get more information about the fraction of invisible loops a section of grain #2 in the 32 dpa sample was imaged using different diffraction vectors. The resulting micrographs are shown in Fig. 6. Comparing the images it is apparent that the contrast of the defects changes due to different loop inclination after tilting the sample to the next diffraction condition. Most of the loops – marked with closed circles – were visible using the $g = \{2\,1\,1\}$ and $g = \{2\,0\,0\}$ diffraction vectors (partially also with $g = \{0\,1\,1\}$). Other loops – numbered 1–3 and marked with dashed circles – were only visible in one of the micrographs. To determine the Burgers vectors of the dislocation loops the invisibility criterion $g \cdot b = 0$ was used. Since the diffraction vectors are not exactly known, there was some ambiguity concerning the determination. The product $g \cdot b$ was calculated for all variants of the used diffraction vectors and $b = \langle 1\,0\,0\rangle, \frac{1}{2}\langle 1\,1\,1\rangle$ to identify the most probable Burgers vectors: $b = \langle 1\,0\,0\rangle$ in case of the loops marked with '1' and $b = \frac{1}{2}\langle 1\,1\,1\rangle$ in all other cases. Corresponding to all visible defects in the three micrographs in Fig. 6 roughly 73% of the defects are visible in case of $g = \{2\,1\,1\}$, 58% in case of $g = \{2\,0\,0\}$ and 15% in case of $g = \{0\,1\,1\}$. These values are not very reliable, because only a very small area of the sample was analyzed by this means so far. A systematic comparison of images taken from a much larger area is needed to reliably determine the ratio of loops with $\langle 1\,0\,0\rangle$ and $\frac{1}{2}\langle 1\,1\,1\rangle$ type Burgers vectors. However, due to invisibility of defects the overall defect densities could be higher than the reported values.

3.3. Effect on hardening

Radiation hardening of RAFM steels was subject of many experimental studies, the mechanical behavior of e.g. EUROFER97 is well known for damage doses up to 70 dpa [1]. The obtained quantitative dataset gives us the possibility to correlate the irradiation induced changes in the microstructure to the changes in the mechanical properties. For this purpose we applied the dispersed barrier hardening model (DBH) [19], which uses simple geometrical considerations to calculate the hardening $\Delta\sigma_x$ caused by the interaction of moving dislocations with randomly distributed obstacles of type x.

$$\Delta\sigma_x = M_T \alpha_x \mu b \sqrt{\sum_{i=1}^{i_{max}} N_i d_i} \qquad (1)$$

According to Eq. (1) the hardening depends on the Taylor factor M_T (3.06 for bcc metals [20]), the obstacle strength of the defects α_x, the

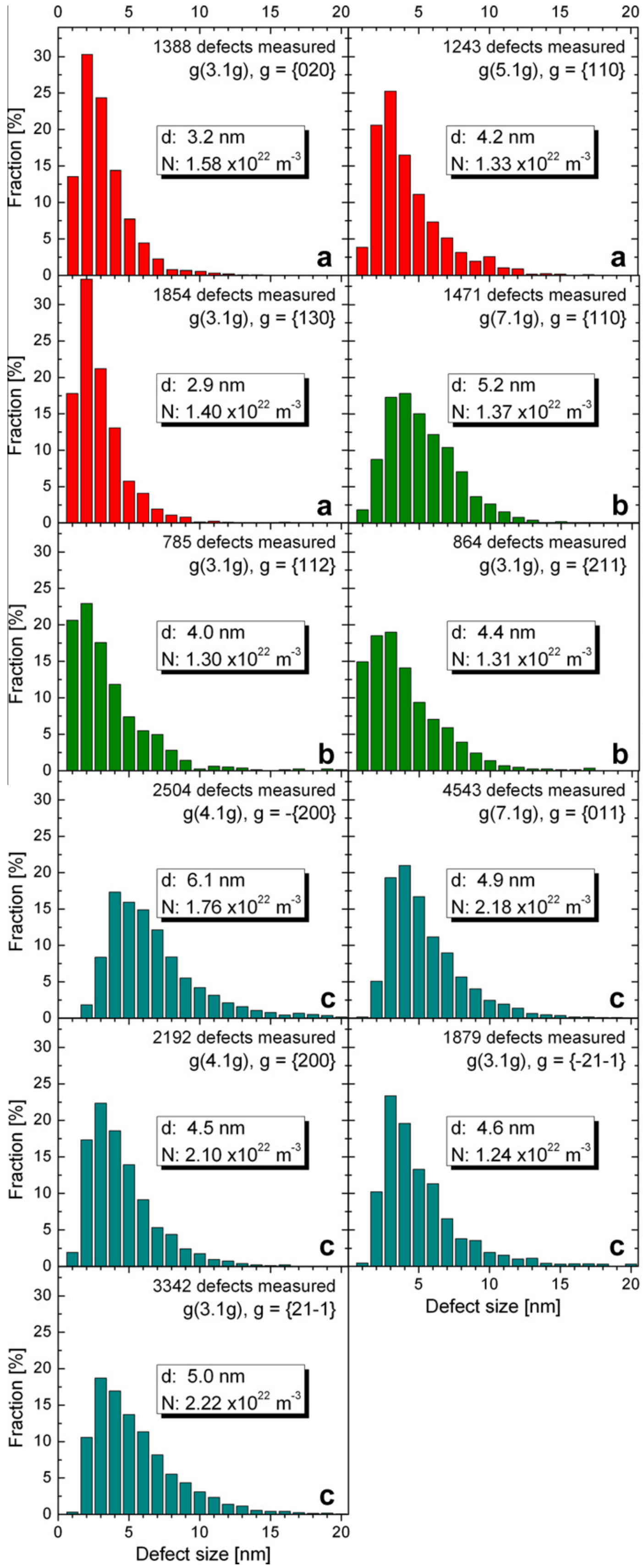

Fig. 5. Size distributions of dislocation loops and defect clusters in EUROFER97 irradiated to: (a) 15 dpa (red), (b) 32 dpa, grain #1 (olive) and (c) 32 dpa, grain #2 (cyan). The underlying number of defects and the corresponding diffraction conditions are indicated in the graphs. Also the average size *d* and volume density *N* of the defects are given in the insets. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Comparison of quantitative microstructural data (dislocation loops & defect clusters) on different irradiated RAFM steels. Only neutron irradiation with an irradiation temperature T_{irr} in the range of 300–330 °C was taken into account.

Material	Irr. source	Dose (dpa)	T_{irr} (°C)	$\langle d \rangle$ (nm)	$\langle N \rangle$ (m^{-3})	Reference
EUROFER97	*n* (BOR 60)	15.0	330	3.4	1.4×10^{22}	This work
EUROFER97	*n* (BOR 60)	32.0	332	4.8	1.7×10^{22}	This work
EUROFER97	*n* (HFR)	16.3	300	14.0	4.0×10^{21}	[15]
F82H	*n* (HFIR)	5.0	300	3.4	1.9×10^{23}	[18]
F82H	*n* (HFR)	8.8	302	5.4	4.5×10^{22}	[17]
F82H	*n* (HFR)	9.2	310	6.9	2.8×10^{22}	[17]

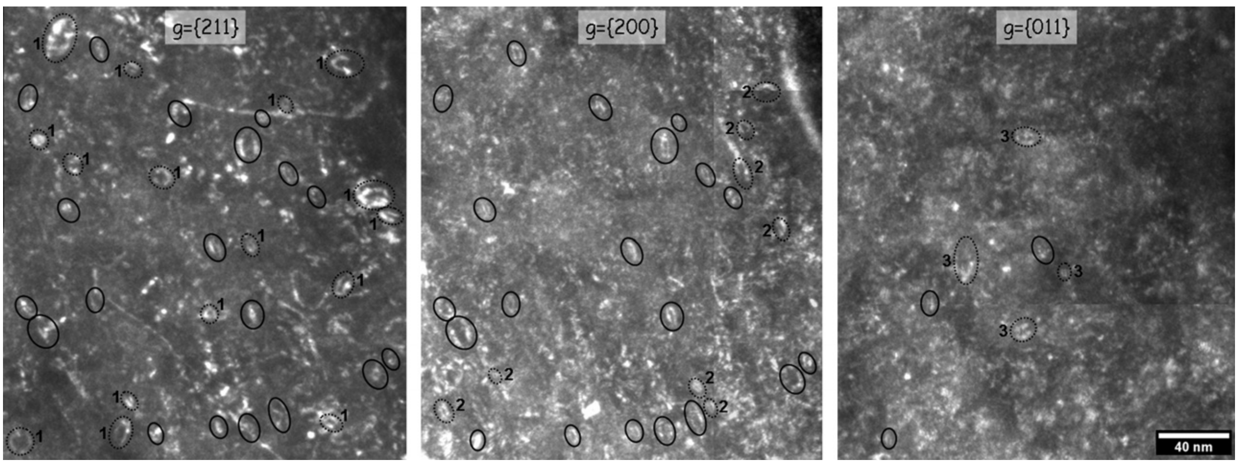

Fig. 6. A section of grain #2 (32 dpa) imaged with 3 different diffraction vectors. The defect contrast changed due to different loop inclination after tilting the sample to the next diffraction condition. Loops visible in several images were marked with solid circles. Loops visible in only one of the micrographs were numbered 1–3 and marked with dashed circles.

shear modulus μ, the magnitude of the Burgers vector b ($\sqrt{3}a_0/2$ for bcc, where a_0 is the lattice parameter [20]) and the distribution of density N_i and diameter d_i of the defects.

The material parameters μ and a_0 were taken from Fe–10Cr model alloys at a temperature of 300 °C, they are 75.3 GPa [21] and 2.88 Å [22], respectively. According to recent molecular dynamics (MD) simulations for the interaction of screw dislocations with small interstitial loops in α-Fe [23], the obstacle strength for loops/clusters at this temperature is about 0.6. Voids were assumed to be rigid obstacles, so their barrier strength is 1. Since a fraction of defects is always invisible in the micrographs (see Section 3.2) it is reasonable to pick the maximum observed defect density N_{max} as input parameter for calculating the hardening. In contrast to the density the sizes of the defects do not depend on their orientation with respect to the electron beam, so using $\langle d \rangle$ is appropriate. These parameters are summarized in Table 2, together with the average *N*- and *d*-values of the voids. Also the resulting hardening values are given. The dose-dependent hardening $\Delta R_{p0.2}$ (0.2% offset yield stress) of EUROFER97 determined by tensile tests [1] is given in Fig. 7. The line reproduced from [1] is according to a model by Whapham and Makin [24], which predicts saturation of hardening at high damage doses. The hardening by loops/clusters calculated with Eq. (1) using the obtained quantitative data in Table 2 is displayed as filled diamonds. The crosses were calculated using $\langle d \rangle$ and $\langle N \rangle$, whereas the error bars represent the experimental scattering of $\Delta\sigma_{loops}$ when each distribution in

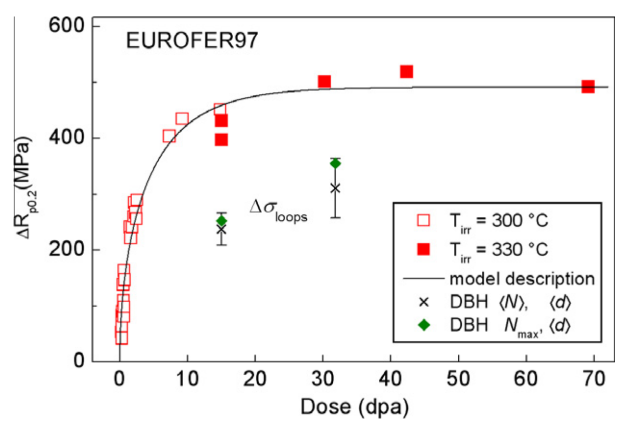

Fig. 7. Based on the obtained quantitative data the hardening by clusters and dislocation loops was estimated using the DBH model. Both the maximum defect density (N_{max}, filled diamonds) and the average one ($\langle N \rangle$, crosses) were used as input parameters. In comparison the dose-dependent hardening $\Delta R_{p0.2}$ of EUROFER97 determined by tensile tests is shown (squares). These data points comprise results from KIT and from literature [26,27]. The solid line is a description according to a model by Whapham and Makin [1,24].

Fig. 5 is used as input parameter. Since the total yield strength increase under irradiation is always a superposition of the effects of different defect populations [25], the yield strength increase

Table 2
Densities and sizes of the two defect types (loops/clusters, voids) and their contribution to hardening, estimated with the dispersed barrier hardening model (DBH).

Dose (dpa)	$N_{loops,max}$ (m^{-3})	$\langle d \rangle_{loops}$ (nm)	$\Delta\sigma_{loops}$ (MPa)	$\langle N \rangle_{voids}$ (m^{-3})	$\langle d \rangle_{voids}$ (nm)	$\Delta\sigma_{voids}$ (MPa)
15.0	1.58×10^{22}	3.4	252	3.6×10^{20}	2.6	56
32.0	2.22×10^{22}	4.8	355	2.3×10^{21}	1.6	110

due to loops alone was considerably lower than the measured $\Delta R_{p0.2}$. According to a recent parametric study by Lambrecht et al. [25], a quadratic superposition law like

$$\Delta\sigma_{tot} = \sqrt{\Delta\sigma_1^2 + \Delta\sigma_2^2 + \ldots} \tag{2}$$

was used to take into account the contribution of voids and other defects, as the corresponding obstacle strengths are comparable. Since the density of voids in EUROFER97 is one (32 dpa) respectively two (15 dpa) orders of magnitude lower than that of the loops, the calculated total hardening $\Delta\sigma_{tot}$ increases only very slightly from 252 to 258 MPa at 15 dpa and from 355 to 372 MPa at 32 dpa when the hardening contributions of voids are included in addition to loops/clusters using Eq. (2) and the data in Table 2. So in our case the contribution of voids can be neglected. There are several possible reasons for the difference (factor $\leqslant 1.8$) between estimated and tensile hardening. Firstly, as mentioned in Section 3.2, a considerable number of loops or clusters are invisible due to their orientation with respect to the electron beam. Secondly certain types of defects, like vacancy clusters or nano-sized precipitates cannot be detected by TEM. Thus studies with different complementary methods like TEM, PAS and SANS are needed to get full quantitative information on radiation damage. Though smaller, defects invisible in the TEM could contribute significantly to hardening because of their high number density. In [28] for example the authors analyzed pure Fe irradiated to 0.2 dpa and reported that the defect densities obtained by SANS and PAS measurements were more than one order of magnitude higher than the density of dislocation loops from TEM investigations. Besides it is well-known that in ferritic/martensitic steels with a chromium content $\geqslant 8\%$ the Cr-rich α' phase precipitates under neutron irradiation [29,30]. Future investigations will show if α' and other radiation induced precipitates can be detected in neutron irradiated EUROFER97 and if these precipitates have a strong impact on hardening.

4. Conclusions

1. The microstructure of EUROFER97 irradiated to 15 dpa and 32 dpa was analyzed quantitatively using different WBDF diffraction conditions. The average size of dislocation loops and defect clusters increased from 3.4 nm to 4.8 nm and the average volume density of the defects from $1.4 \times 10^{22}\ m^{-3}$ at 15 dpa to $1.7 \times 10^{22}\ m^{-3}$ at 32 dpa.
2. Voids were identified by through-focus series, their average size was 2.6 nm at 15 dpa and 1.6 nm at 32 dpa. The corresponding volume densities were $3.6 \times 10^{20}\ m^{-3}$ and $2.3 \times 10^{21}\ m^{-3}$, respectively.
3. Majority of the dislocation loops had Burgers vectors of the $\frac{1}{2}\langle 1\,1\,1\rangle$ type.
4. Based on the obtained quantitative data the irradiation-induced hardening was estimated with the DBH model. The calculated hardening was significantly smaller than determined in the post-irradiation tensile tests, which shows that additional complementary methods like PAS and SANS are needed to get full quantitative information on radiation damage.

Acknowledgments

This work, supported by the European Communities under the contract of Association between EURATOM and Karlsruhe Institute of Technology, was carried out within the framework of the European Fusion Development Agreement. The views and opinions expressed herein do not necessarily reflect those of the European Commission. This work is partly funded by the Helmholtz Association and the Russian Foundation for Basic Research under the Grant HRJRG-013.

References

[1] E. Gaganidze, C. Petersen, E. Materna-Morris, C. Dethloff, O.J. Weiß, J. Aktaa, A. Povstyanko, A. Fedoseev, O. Makarov, V. Prokhorov, J. Nucl. Mater. 417 (2011) 93–98.
[2] E. Gaganidze, H.-C. Schneider, B. Dafferner, J. Aktaa, J. Nucl. Mater. 355 (2006) 83–88.
[3] C. Petersen, J. Aktaa, E. Diegele, E. Gaganidze, R. Lässer, E. Lucon, E. Materna-Morris, A. Möslang, A. Povstyanko, V. Prokhorov, J. Rensman, B. van der Schaaf, H.-C. Schneider, 21st IAEA Fusion Energy Conf., Chengdu, China, October 16–21, FT/1-4Ra, 2006.
[4] C. Petersen, V. Shamardin, A. Fedoseev, G. Shimansky, V. Efimov, J. Rensman, J. Nucl. Mater. 307–311 (2002) 1655–1659.
[5] D.J.H. Cockayne, I.L.F. Ray, M.J. Whelan, Phil. Mag. 20 (1969) 1265–1270.
[6] Y. Tomokiyo, S. Matsumura, T. Manabe, J Microscopy 194 (1999) 210–218.
[7] M.L. Jenkins, J. Nucl. Mater. 216 (1994) 124–156.
[8] O.J. Weiß, E. Gaganidze, J. Aktaa, Adv. Sci. Technol. 73 (2010) 118–123.
[9] J. Marian, B.D. Wirth, R. Schäublin, J.M. Perlado, T. Diaz de la Rubia, J. Nucl. Mater. 307–311 (2002) 871–875.
[10] T. Byun, N. Hashimoto, K. Farrell, E. Lee, J. Nucl. Mater. 349 (2006) 251–264.
[11] ImageJ, Image Processing and Analysis in Java, 2011 <http://rsbweb.nih.gov/ij/>.
[12] P.M. Kelly, A. Jostsons, R.G. Blake, J.G. Napier, Phys. Status Solidi (a) 31 (1975) 771–780.
[13] S.M. Allen, Phil. Mag. A 43 (1981) 325–335.
[14] V. Hou, Thickness by CBED, Digital Micrograph(tm) Script Database, 2011 <http://www.felmi-zfe.tugraz.at/dm_scripts/>.
[15] M. Klimenkov, E. Materna-Morris, A. Möslang, J. Nucl. Mater. 417 (2011) 124–126.
[16] M. Klimenkov, Private Communication, 2011.
[17] R. Schäublin, M. Victoria, in: G. Lucas, L. Snead, M. Kirk Jr., R. Elliman (Eds.), Microstructural Processes in Irradiated Materials, MRS Symp. Proc., MRS, vol. 650, pp. R181–R186.
[18] H. Tanigawa, R.L. Klueh, N. Hashimoto, M.A. Sokolov, J. Nucl. Mater. 386–388 (2009) 231–235.
[19] A. Seeger, in: Proc. 2nd UN Int. Conf. on Peaceful Uses of Atomic Energy, vol. 6, p. 205.
[20] N. Hashimoto, T. Byun, K. Farrell, S. Zinkle, J. Nucl. Mater. 336 (2005) 225–232.
[21] G. Speich, A. Schwoeble, W. Leslie, Metall. Mater. Trans. B 3 (1972) 2031–2037.
[22] D. Terentyev, N. Juslin, K. Nordlund, N. Sandberg, J. Appl. Phys. 105 (2009) 103509–103512.
[23] D. Terentyev, D.J. Bacon, Y. Osetsky, Phil. Mag. 90 (2010) 1019–1033.
[24] A.D. Whapham, M.J. Makin, Phil. Mag. 5 (1960) 237–250.
[25] M. Lambrecht, E. Meslin, L. Malerba, M. Hernández-Mayoral, F. Bergner, P. Pareige, B. Radiguet, A. Almazouzi, J. Nucl. Mater. 406 (2010) 84–89.
[26] A. Alamo, J. Bertin, V. Shamardin, P. Wident, J. Nucl. Mater. 367–370 (2007) 54–59.
[27] E. Lucon, R. Chaouadi, M. Decreton, J. Nucl. Mater. 329–333 (2004) 1078–1082.
[28] E. Meslin, M. Lambrecht, M. Hernández-Mayoral, F. Bergner, L. Malerba, P. Pareige, B. Radiguet, A. Barbu, D. Gómez-Briceño, A. Ulbricht, A. Almazouzi, J. Nucl. Mater. 406 (2010) 73–83.
[29] M.H. Mathon, Y. Carlan, G. Geoffroy, X. Averty, A. Alamo, C.H. Novion, J. Nucl. Mater. 312 (2003) 236–248.
[30] E. Wakai, Y. Miwa, N. Hashimoto, J. Robertson, R. Klueh, K. Shiba, K. Abiko, S. Furuno, S. Jitsukawa, J. Nucl. Mater. 307–311 (2002) 203–211.